MOHAN KUMAR B.H

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EDUCATION

Bachelor of Technology in Computer Science and Engineering (AIML) **2022 - 2026**
Sai Vidya Institute of Technology, Rajanukunte
CGPA: 8.80
Relevant Coursework: Machine Learning, Deep Learning, Prompt Engineering, RAG, LLM.

TECHNICAL SKILLS

1. **Programming Languages:** Python (Proficient), SQL, C/C++, React, JavaScript.
2. **Core CS:** Data Structures & Algorithms, OOP, OS, DBMS, Computer Networks, System Design.
3. **AI Expertise:** Prompt Engineering, LLM, Transformers, RAG, Fine-Tuning, Agentic AI, Generative AI.
4. **Machine Learning & Frameworks:** TensorFlow, PyTorch, Scikitlearn, OpenCV, Keras, HuggingFace.
5. **Tools & Cloud:** Git & GitHub, Docker, Power BI, AWS, Linux, Claude Code, Copilot, Antigravity.

EXPERIENCE

DIGITALTRANSOLS AI PRIVATE LIMITED, Bangalore **Feb 2026 – June 2026**
AI Intern

- AI Interview Engine (FastAPI, Next.js, Gemini 2.0 Flash): Developed a real-time, bilingual (Kannada/English) platform with multimodal audio transcription and adaptive, rubric-based follow-up questioning to generate detailed performance reports.
- Bulk-Recruitment Pipeline (Telegram): Built a concurrent automated system to parse CSVs, screen candidate profiles against job criteria, and orchestrate interview invitations.

PROJECTS

1. **Intelli-Eye: ML-Based Surveillance System** **Aug 2025 – Dec-2025**
   - **Languages & Frameworks:** Python, OpenCV, YOLOv8.
   - **Deep Learning & Hardware:** GPU-Accelerated Pipelines, CNNs, Edge Deployment.
   - **Signal Processing:** Audio Anomaly Detection (Screams, Glass Breaking).
   - **Infrastructure**: Real-time Alert Engines, Smart City Frameworks.
2. **Personalized Music Recommendation System** **Dec 2024 – Feb 2025**
   - **Machine Learning:** Collaborative Filtering, Content-Based Filtering, K-Means Clustering.
   - **Data Processing:** Python (Pandas, NumPy, Scikit-learn), Spotify API.
   - **Mathematics & Stats:** PCA (Dimensionality Reduction), t-SNE, Cosine Similarity.
   - **Visualization:** Matplotlib, Seaborn, Plotly.
3. **Stock Portfolio analysis Agent** **June 2026 – Aug 2026**
   - **CrewAI + FastAPI:** Orchestrates the multi-step stock-analysis workflow and exposes it via a high-performance asynchronous API.
   - **React + Next.js:** Provides a responsive, server-rendered front-end where users input prompts and view live streaming results.
   - **CopilotKit AG - UI Protocol:** Streams agent events (tool calls, intermediate data, final insights) from the backend to the UI in real time via WebSockets.
   - **yfinance + pandas:** Retrieves historical market data and performs the financial calculations, data wrangling, and chart generation.

PUBLICATION

Paper title: Intelli-Eye: A Holistic Deep-Learning Architecture for Predictive Threat Analysis and Intelligent Alert Automation.
Conference: 2026 International Conference on Natural Language Processing and Computer Vision (ICNPCV), IEEE Bangalore.